

P. NITHIN EMMANUEL

Python Developer | AI/ML Engineer

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PROFESSIONAL SUMMARY

GenAI Engineer / Python Developer with production experience building RAG pipelines, Text2SQL systems, and LLM query-safety layers using LangChain, LlamaIndex, and Milvus, backed by Flask/FastAPI services on an enterprise AI platform. Skilled in async data flows and Python-based test automation. Comfortable owning a feature end-to-end, from technical design through implementation and validation.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, Flask, FastAPI, REST API design, async task workers

GenAI / LLM: RAG, LangChain, LlamaIndex, Text2SQL/NL2SQL, LLM prompt engineering, hybrid retrieval

Vector & Data: Milvus, BAAI/bge-large-en-v1.5 embeddings, custom node parsers/splitters

Testing & Tools: pytest, CSV/Excel-driven test automation, Docker, Git, JIRA, Postman

PROFESSIONAL EXPERIENCE

Python Developer — Solix Technologies

IDA-2 (Intelligent Data Assistant) Platform | 2024 – Present

- **Enabled natural-language access to governed enterprise data** — evidenced by production integration with the platform's query guardrails/AST validation layer — by building a Text2SQL pipeline using LangChain and LlamaIndex orchestration to translate natural-language questions into executable SQL over the governed data model.
- **Stood up production-grade RAG retrieval infrastructure** — evidenced by an authenticated, Dockerized Milvus deployment serving 1024-dimension embeddings — by integrating LlamaIndex with BAAI/bge-large-en-v1.5 embeddings and building custom document node parsers/splitters plus a regex-based metadata extraction POC.
- **Closed a critical query-safety gap in the platform's LLM query pipeline** — evidenced by root-causing and resolving a BOKG boundary-breach defect — by developing an AST-based query validation layer (L1) and temporarily owning the downstream Guardrails (L2) layer.
- **Delivered an auditable, concurrency-safe consumption-billing engine** spanning 27 database tables — evidenced by zero ledger-consistency defects — by designing an atomic, append-only ledger with SELECT FOR UPDATE concurrency control and a 4-tier rate resolution strategy, exposed via Flask/FastAPI REST APIs for rate lookup, debit calculation, and account provisioning.
- **Increased regression coverage across the platform's AI pipeline stages** — evidenced by a 104-case suite spanning 7 ambiguity scenarios — by building 7 pytest-based, data-driven (CSV/Excel) API automation frameworks (Dev1–Dev7) and a Google Apps Script dashboard for cross-pipeline visualization.
- **Enabled accurate LLM cost monitoring** — evidenced by thread-safe input/output token accounting — by implementing a double-checked-locking TokenCalculator utility.
- **Reduced cross-team ambiguity on sprint requirements** — evidenced by finalized API contracts and data models for sprint deliverables — by authoring system design documentation for the platform's Data Ask module (RAG, hybrid retrieval).
- **Accelerated onboarding of 132 Oracle EBS tables** into an AI-enriched metadata catalog — evidenced by structured documentation across 10 modules (AP, AR, BEN, CE, GL, HR, INV, OM, PAY, PO) — by producing consultant-grade business-context content combining AI-authored sections with extracted column descriptions.

EDUCATION

M.Tech, Computer Science

Vaagdevi College of Engineering | 2024 – 2026 (Expected)

Bachelor of Technology, Cyber Security

Institute of Aeronautical Engineering (IARE), Hyderabad | June 2020 – June 2024

CERTIFICATIONS

- Python
- Advanced SQL
- Advanced Cyber Security — Threats & Governance